COS 30049 Computing Technology Innovation Project

Week 6: How a Model Learns

Regression and Clustering, one step at a time

Acknowledgement of Country

We respectfully acknowledge the Wurundjeri People of the Kulin Nation, who are the Traditional Owners of the land on which Swinburne's Australian campuses are located in Melbourne's east and outer-east, and pay our respect to their Elders past, present and emerging.

We are honoured to recognise our connection to Wurundjeri Country, history, culture, and spirituality through these locations, and strive to ensure that we operate in a manner that respects and honours the Elders and Ancestors of these lands.

We also respectfully acknowledge Swinburne's Aboriginal and Torres Strait Islander staff, students, alumni, partners and visitors.

We also acknowledge and respect the Traditional Owners of lands across Australia, their Elders, Ancestors, cultures, and heritage, and recognise the continuing sovereignties of all Aboriginal and Torres Strait Islander Nations.



Objectives

1. **Understand what it means for a model to learn from data**

We build linear regression from nothing, then name each piece with its formula

2. **Know which regression model to reach for, and why**
3. **Follow K-means one iteration at a time, with the arithmetic written out**
4. **Know why a model can fit the training data perfectly and still be useless**

Next week builds on this with classification, which is the task Assignment 2 asks for. This week is about the machinery underneath it.

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What machine learning actually is

and what it is not

Ordinary programming: you write the rules, the computer applies them.

Machine learning: you supply examples, the computer works out the rules.

That is the whole idea. Everything after this is detail about how the working out is done, and how you check whether the rules it found are any good.

Writing the rules yourself

if length > 50 and has_link:

    flag as spam

Letting the model find them

here are 5000 labelled messages,

work out what separates them

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Two different questions

Supervised

You have the answers.

Every example comes with a label. The model learns to reproduce it on examples it has not seen.

Today: regression
Next week: classification

Unsupervised

You have no answers.

No labels at all. The model looks for structure that is already there but that nobody wrote down.

Today: K-means clustering

Assignment 2 asks for both: a classifier that detects your target, and a clustering step that finds structure the label column does not contain.

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Part 1

Linear regression, built from nothing

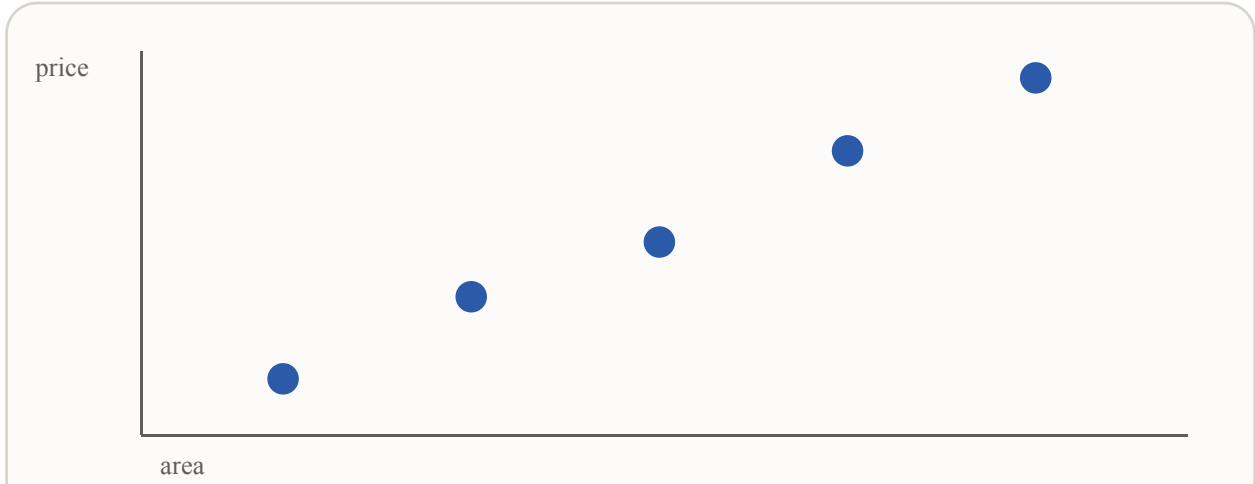
Six steps. No formulas to memorise.

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Step 1: here is the data, there is no model yet

Five houses. Area in square metres, price in thousands.

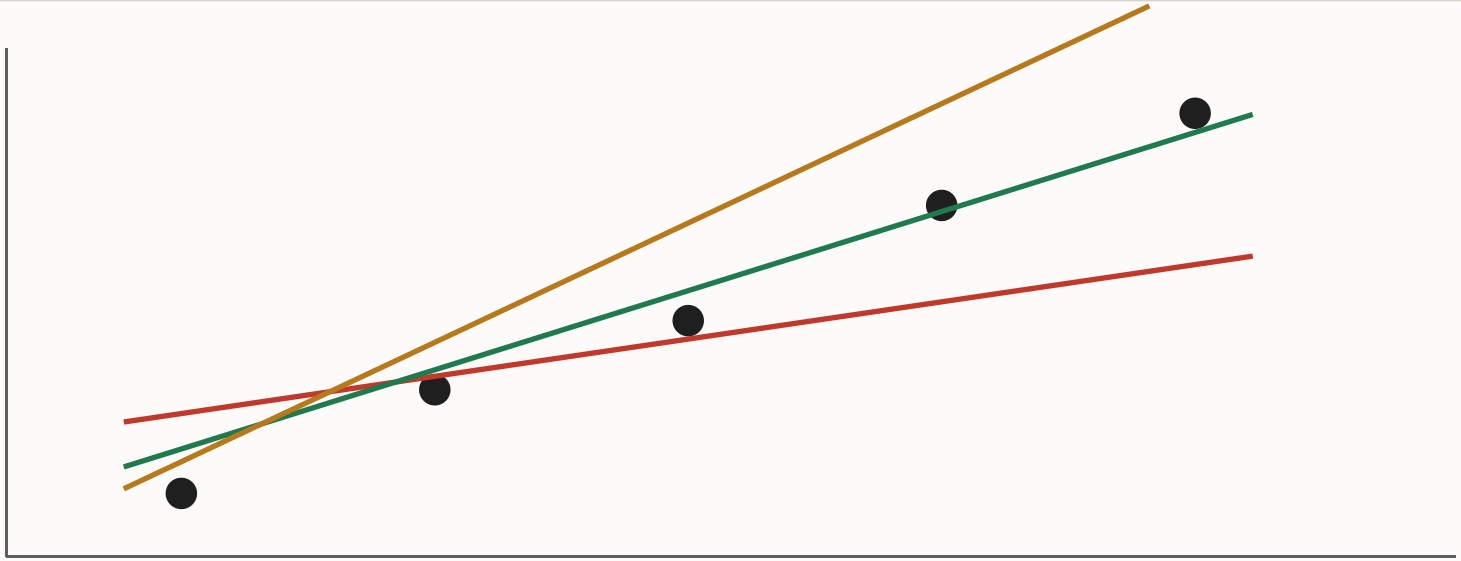
area	price
50	310
70	400
90	460
110	560
130	640



Question for the room: what would you charge for a house of 120 square metres? You already have an instinct. The rest of this section turns it into arithmetic.

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Step 2: guess three lines, and argue about them



C: too steep

B: about right

A: too low

Everyone picks B. The important question is WHY. The answer is always some version of 'it passes closer to the dots', and that sentence is the whole of machine learning.

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Step 3: turn 'closer to the dots' into a number

For each house, how far is the line from the real price?

area	real price	line says	error
50	310	372	-62
70	400	442	-42
90	460	512	-52
110	560	582	-22
130	640	652	-12

The error of one house

$$e_i = y_i - \hat{y}_i$$

real price minus what the line said

Why not just add them up?

-62 -42 -52 -22 -12 = -190

A line too high in some places and too low in others has its errors cancel out to zero.

The hat on y-hat is the convention for what the model predicted, as opposed to plain y which is the truth. You will meet it in every textbook from here on.

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Step 4: square the errors

This is where mean squared error comes from, and why it has that name

area	error	error squared
50	-62	3844
70	-42	1764
90	-52	2704
110	-22	484
130	-12	144

Squaring does two jobs

It removes the sign, so errors cannot cancel each other out.

And it punishes a big miss far more than a small one: being out by 60 costs 3600, while being out by 12 costs only 144.

The average of those squares is the mean squared error.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

our five houses:

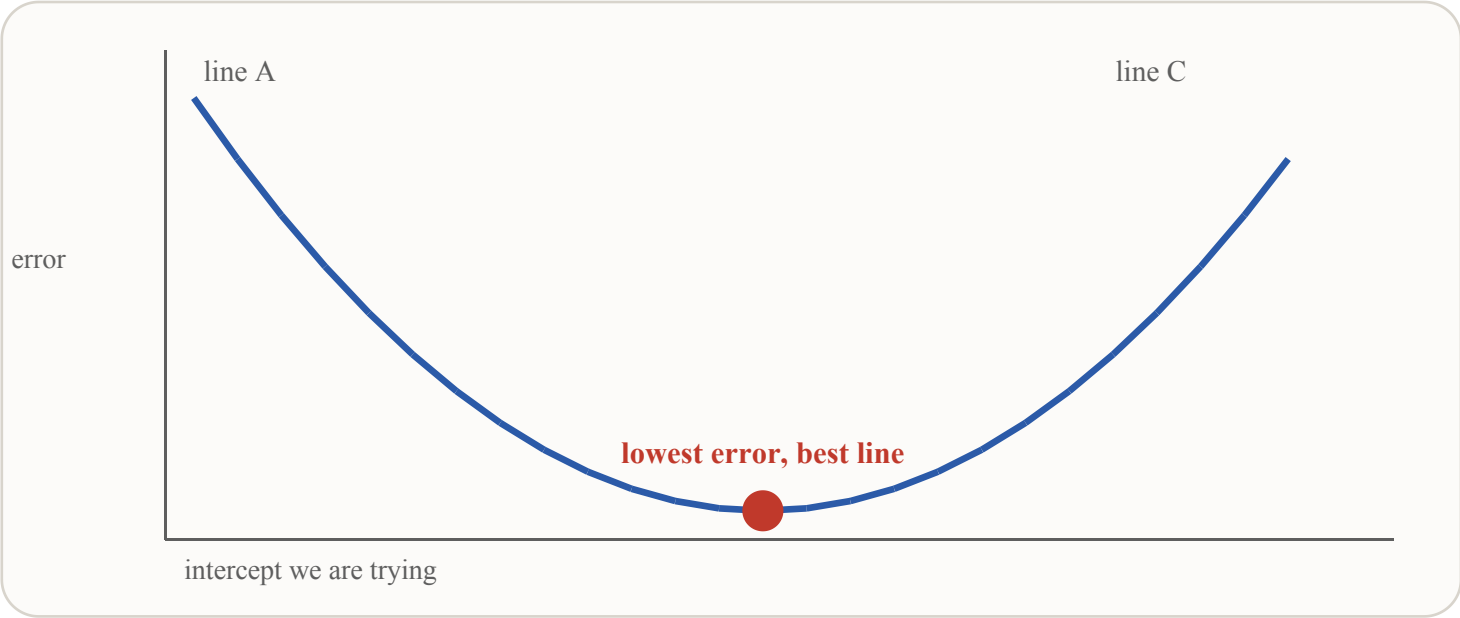
$$MSE = \frac{(-62)^2 + (-42)^2 + (-52)^2 + (-22)^2 + (-12)^2}{5} = \frac{8940}{5} = 1788$$

Read the formula against the table. The sigma is the total row, the n is 5, and the bracket squared is the last column. Nothing in it is new.

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Step 5: try every line and plot the score

Keep the slope fixed, slide the intercept up and down



This is the most useful picture in the unit. Error is not a fixed property of the data; it is a function of the numbers you chose. Training means walking downhill on this curve.

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Step 6: that was training

- A computer does exactly what we just did, with two differences.
- It adjusts the slope and the intercept together, so the curve becomes a bowl in three
- dimensions rather than a line in two. And it does not try every value: it works out which
- direction is downhill and steps that way, until the error stops falling.

One step of gradient descent

$$b_1 \leftarrow b_1 - \alpha \frac{\partial\, MSE}{\partial\, b_1}$$

The derivative on the right is the slope of the curve you just saw. It says which way is downhill, and how steep it is there.

Alpha is the learning rate, the size of each step. Too small and training crawls; too large and it overshoots the bottom.

In code all six steps collapse into one line: `model.fit(X, y)`

Everything we just did by hand is what that line is doing.

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Three ways to measure the same error

They differ only in how harshly they treat a big miss

MSE

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Squares every error, then averages. A miss of 60 costs 25 times more than a miss of 12.

Use when large errors are especially costly.

our five houses: 1788

RMSE

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

The square root of MSE, which puts the number back into the units of the thing you predicted.

Use when you want to say out by about X thousand.

our five houses: 42.3

MAE

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Takes the size of each error and averages, with no squaring at all. Every miss counts equally.

Use when outliers should not dominate the score.

our five houses: 38.0

All three describe the same set of mistakes. Choosing between them is a statement about which mistakes you care about, and the report has to justify it.

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Fitting the data perfectly is not the goal

Underfitting

The model is too simple to follow the pattern that is really there.

Bad on training data, bad on new data.

About right

Follows the pattern, ignores the noise around it.

Good on training data, good on new data.

Overfitting

Bends to hit every single training point, noise included.

Perfect on training data, poor on new data.

This is why the test set exists, and why you must not look at it while choosing your model. A score on data the model has already seen tells you nothing about whether it will work.

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One more score: R squared

The others ask how far off you are. This one asks whether you beat a lazy guess.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

the same thing in words

$$R^2 = 1 - \frac{\text{error of my model}}{\text{error of always guessing the average}}$$

1.0	perfect, the model explains everything
0.0	no better than always guessing the average price
below 0	worse than guessing the average, which does happen

On the wine data in this week's seminar, alcohol alone predicts quality with R squared 0.185. All eleven measurements give 0.351. Neither is near 1, and that is honest: taste is not fully explained by eleven chemical numbers.

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The same model, with more flexibility

Polynomial regression is linear regression with extra columns added

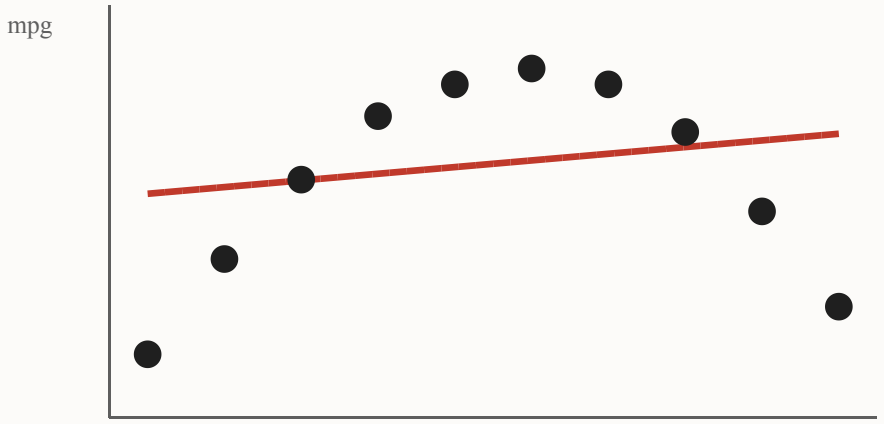
degree 1	$\hat{y} = b_0 + b_1 x$	a straight line
degree 2	$\hat{y} = b_0 + b_1 x + b_2 x^2$	one bend
degree 3	$\hat{y} = b_0 + b_1 x + b_2 x^2 + b_3 x^3$	two bends
degree d	$\hat{y} = b_0 + b_1 x + b_2 x^2 + \ldots + b_d x^d$	d minus one bends

The model is still linear in the coefficients b. Only the columns you feed it changed, which is why sklearn implements this by transforming the features and then calling ordinary linear regression.

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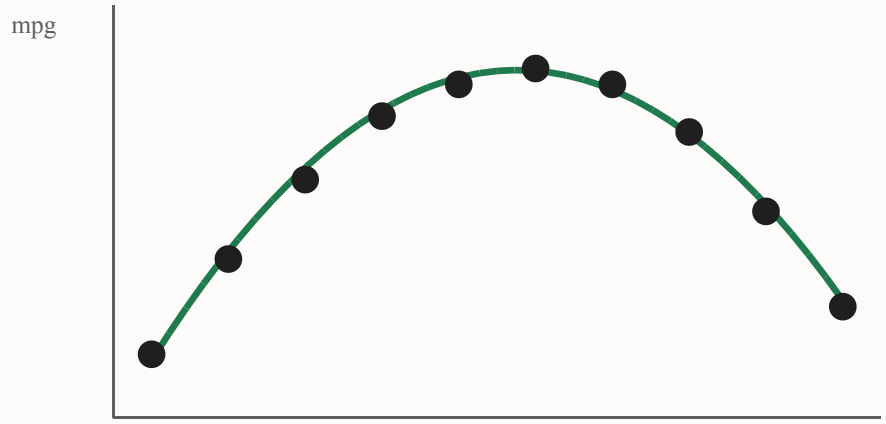
What polynomial regression is for

Fuel economy against speed. Too slow wastes fuel, and so does too fast.



a straight line

cannot bend, so it fits nothing R squared 0.041



add one squared term

follows the shape R squared 0.996

$$\hat{y} = b_0 + b_1 x + b_2 x^2$$

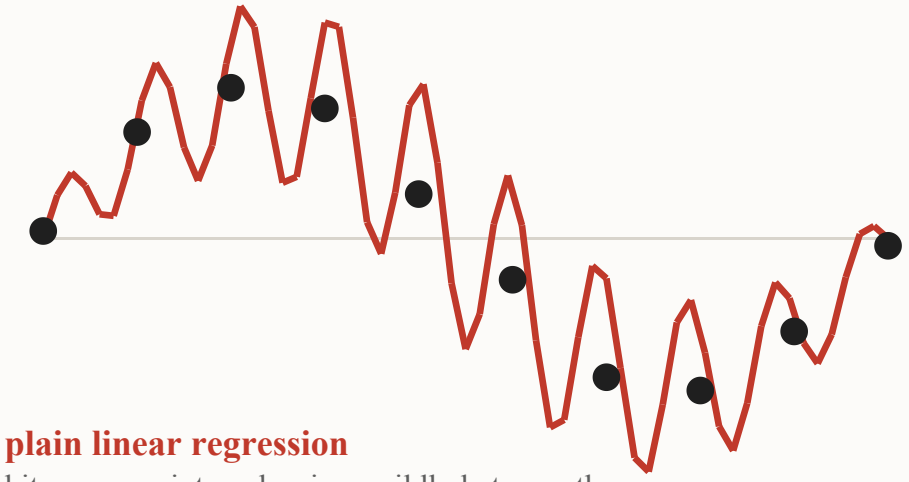
One extra column, x squared, is all that changed. The model is still linear regression; it just has more to work with.

This is the honest reason to reach for polynomial regression: you looked at the scatter plot, saw a bend, and the straight line could not follow it.

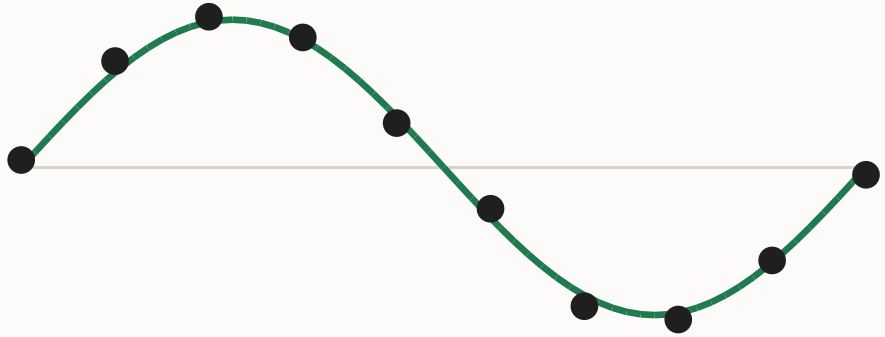
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What ridge regression is for

A degree 9 curve through 10 points. Flexibility with nothing holding it back.



plain linear regression
hits every point, and swings wildly between them



the same fit, with ridge
misses a few points, follows the real shape

	plain	ridge	
largest coefficient	99,665	114	shrunk by a factor of 800
error on the training points	0.0000	0.0084	slightly worse, as expected
error against the true curve	0.0500	0.0129	four times better

Ridge adds a penalty for large coefficients, so the fit has to justify every bit of wiggle it uses. It gives up a little accuracy on the training points and buys a lot on everything else.

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So which regression model do you reach for?

Start at the top. Only move down when the row above genuinely fails.

Linear regression

Use when: the relationship looks straight, or you have no reason to think otherwise

Start here every time. It is fast, and every coefficient means something you can say out loud.

Polynomial regression

Use when: the scatter plot bends, and the bend is real rather than noise

Degree 2 or 3 is almost always enough. Above that you are fitting noise, as the next slide shows.

Ridge regression

Use when: you have many features, several measuring nearly the same thing

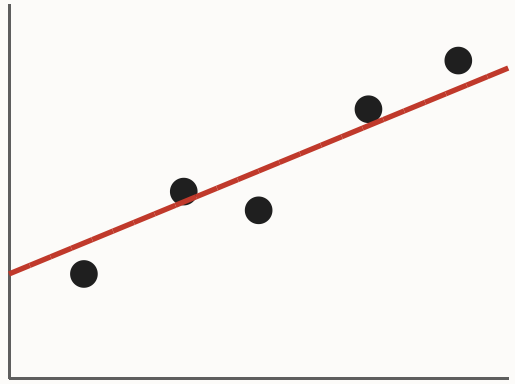
Adds a penalty on large coefficients, so no single feature can dominate the fit.

The honest default is the first row. Reaching for a fancier model before you have shown the simple one failing is the most common mistake in a first machine learning report.

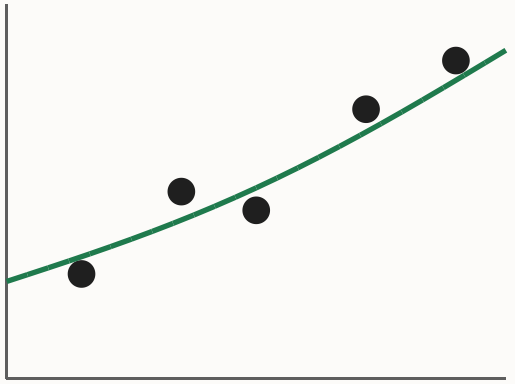
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Overfitting, made visible

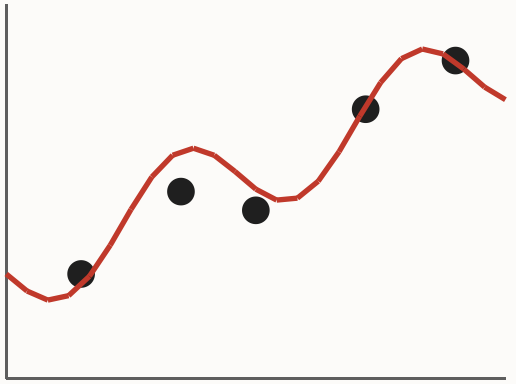
The same five points, fitted with curves of increasing flexibility



degree 1
underfits: misses the curve



degree 3
about right



degree 10
overfits: hits every point

The third curve passes through every point, so its training MSE is zero. It is also the one you would least want to use on a house it has never seen.

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Part 2

K-means: when nobody gives you the answers

Four steps, repeated until nothing moves.

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A different question

Regression asks 'what number'. Clustering asks 'which of these belong together'.

There is no label column. Nothing tells you the right answer, so nothing can tell you your accuracy either.

What you get instead is a grouping, and the work is explaining what each group has in common. A cluster you cannot describe is not a result.

Assignment 2 asks you to cluster within a single class: only the phishing URLs, or only the injection prompts. Every cluster then carries the same label, so the label tells you nothing and you are forced to describe the group by what is actually inside it. That description is what earns the marks.

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K-means, four steps

1. **Pick k, then drop k markers at random**
   You choose how many groups you want. The markers start in arbitrary places.

2. **Assign every point to its nearest marker**
   That is the only rule. Nearest by straight-line distance.

3. **Move each marker to the middle of its group**
   The marker relocates to the average position of the points that chose it.

4. **Repeat steps 2 and 3**
   Points change allegiance, markers shift again. Stop when nothing moves.

Draw this on the board with a dozen dots and two markers. Three rounds is enough for the class to watch it settle, and watching it settle is the point.

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Step 2 in detail: which centroid is nearest?

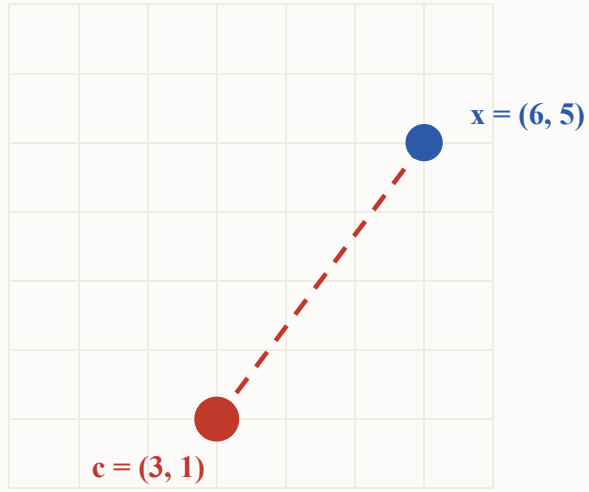
Straight-line distance, the same formula you met in high school geometry

$$d(x, c) = \sqrt{(x_1 - c_1)^2 + (x_2 - c_2)^2}$$

a worked example

`point x = (6, 5)` `centroid c = (3, 1)`

$$d = \sqrt{(6-3)^2 + (5-1)^2} = \sqrt{9+16} = \sqrt{25} = 5$$



Repeat that for every point against every centroid. With 1500 rows and k of 3 that is 4500 distances per round, which is why this is a job for a computer and not for us.

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Step 3 in detail: move the centroid

The new position is the average of the points that chose it

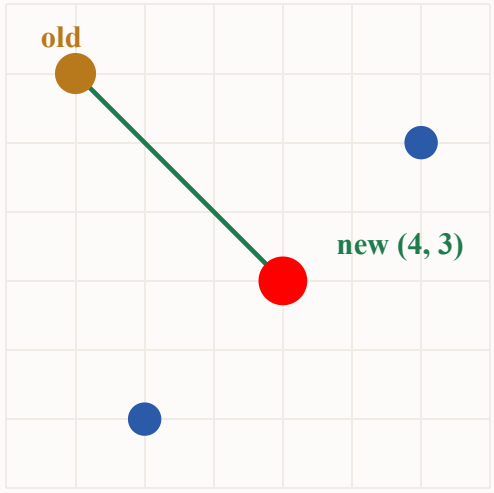
$$c_k = \frac{1}{|S_k|} \sum_{x \in S_k} x$$

Add up the points in the group, then divide by how many there are.

three points chose this centroid

`(2, 1)` `(4, 3)` `(6, 5)`

$$c = \left(\frac{2+4+6}{3}, \frac{1+3+5}{3} \right) = (4,\ 3)$$

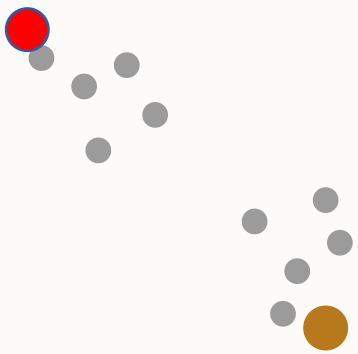


The centroid does not have to land on one of your data points, and usually will not. It is the balance point of the group.

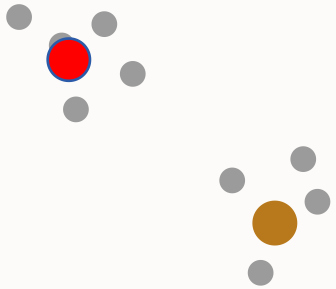
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Step 4: watch it settle

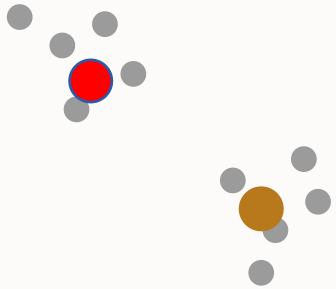
Same points, three rounds. The centroids move less each time.



start
centroids dropped at random



after round 1
each moved toward its group



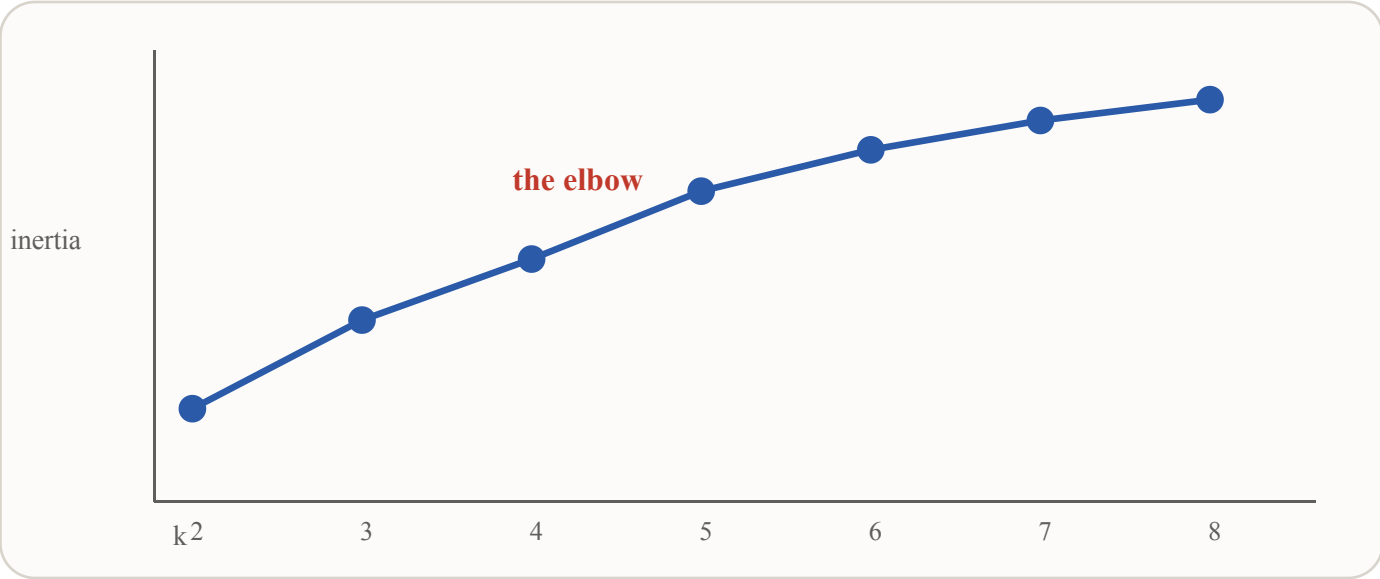
after round 2
barely moved, so we stop

Stopping is not a fixed number of rounds. K-means stops when the assignments stop changing, which on real data usually takes fewer than twenty.

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Choosing k, part one: the elbow

The algorithm cannot tell you how many groups exist. You decide.



Inertia always falls as k rises, so you cannot simply minimise it. Look for the bend, where another cluster stops buying you much. On the wine data the bend is soft, and saying so is itself a finding.

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Choosing k, part two: the silhouette score

A second opinion, and one that can be compared fairly across values of k

$$s = \frac{b - a}{\max(a,\, b)}$$

in words, for a single point

$$s = \frac{\text{distance to nearest other cluster} \;-\; \text{distance within my own}}{\text{whichever of those is larger}}$$

close to 1	the point sits comfortably inside its own cluster
around 0	the point is on the boundary and could go either way
below 0	the point is closer to another cluster than to its own

Average it over every point and you get one number for the whole clustering.

Unlike inertia, silhouette does not automatically improve as k rises, so the highest score is a genuine recommendation. On the wine data every k scores below 0.25, which tells you the groups overlap heavily.

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Judging a clustering when there is no right answer

Inertia

How tightly packed each cluster is.

Always falls as k rises, so it can only be read as a curve, never as a single number.

Silhouette score

Compares how close a point is to its own cluster against the nearest other cluster. Runs from -1 to 1.

Above 0.5 is strong structure. Around 0.2 means heavy overlap.

Read the members

Take a handful of rows from each cluster and read them.

If you cannot say what the group has in common, the number does not matter.

The third box is the one Assignment 2 marks. A silhouette score on its own is not a finding.

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What to take away

Training is downhill walking

You define what error means, then adjust the numbers until it stops falling.

Perfect on training data is a warning, not an achievement

The test set is the only honest measure of whether a model will work.

Clustering has no accuracy

Only a grouping, and your explanation of what each group contains.

Next week: classification, and how to tell whether a model is any good.

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